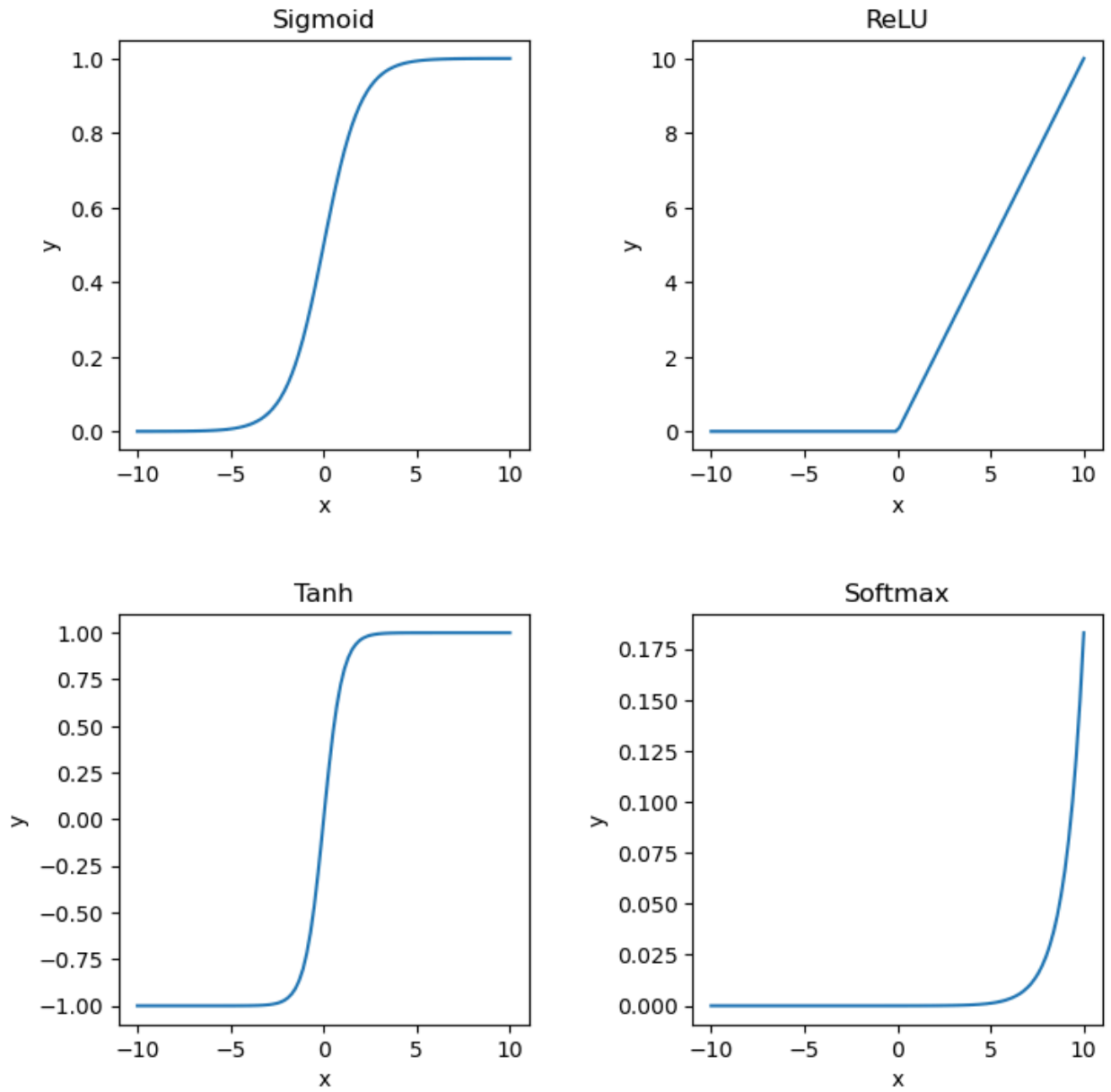




```
In [ ]: PAWAR SAKSHI MAHENDRA  
        ROLL NO - 49  
        PRACTICAL NO - 02
```

```
In [8]: import numpy as np  
import matplotlib.pyplot as plt  
  
def sigmoid(x):  
    return 1 / (1 + np.exp(-x))  
  
def relu(x):  
    return np.maximum(0, x)  
  
def tanh(x):  
    return np.tanh(x)  
  
def softmax(x):  
    e_x = np.exp(x - np.max(x)) # for numerical stability  
    return e_x / e_x.sum()  
  
# Create x values  
x = np.linspace(-10, 10, 100)  
  
# Create plots  
fig, axs = plt.subplots(2, 2, figsize=(8, 8))  
  
axs[0, 0].plot(x, sigmoid(x))  
axs[0, 0].set_title('Sigmoid')  
  
axs[0, 1].plot(x, relu(x))  
axs[0, 1].set_title('ReLU')  
  
axs[1, 0].plot(x, tanh(x))  
axs[1, 0].set_title('Tanh')  
  
axs[1, 1].plot(x, softmax(x))  
axs[1, 1].set_title('Softmax')  
  
# Title  
fig.suptitle('Common Activation Functions')  
  
# Correct loop  
for ax in axs.flat:  
    ax.set(xlabel='x', ylabel='y')  
  
# Spacing  
plt.subplots_adjust(left=0.1, bottom=0.1, right=0.9, top=0.9, wspace=0.4, hspa  
plt.show()
```

Common Activation Functions



In []: